

Karpaga Vinayaga College of Engineering and Technology

Project Review - II

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MCA 2 YEAR**



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Position: Junior Software Developer

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Project Title: Automated Question Bank Generation System using LLM and Generative AI

ABSTRACT

- In today's AI-driven educational landscape, traditional methods of question creation fall short in addressing the growing demand for scalable, diverse, and pedagogically sound assessment content. This paper introduces an AI-powered Automated Question Bank Generation System that leverages the rapid advancements in Large Language Models (LLMs) and Generative AI to streamline and enhance question development for academic and professional examinations [1], [2]. By integrating cutting-edge Natural Language Processing (NLP) techniques such as BERT [3] and the Transformer architecture [4], the system generates high-quality, contextually relevant questions across multiple subjects and difficulty levels. The architecture supports various question formats—including multiple-choice questions (MCQs), true/false, short answer, and descriptive types—ensuring adaptability to diverse assessment requirements [5]. Unlike conventional, rule-based question-authoring tools, this generative approach automates the creation process while maintaining fairness and quality across a broad content spectrum [7]. Additionally, the system provides educators with intuitive interfaces and customizable templates, significantly reducing manual effort and promoting efficient content deployment [8]. Our contributions include the design of a generative question bank framework, integration of automated quality assurance modules, and validation of the system's effectiveness in producing diverse and standards-compliant assessment material. Limitations include sensitivity to training data quality, challenges in generating domain-specific or technical diagrams, and constraints in symbolic reasoning. Future directions involve adaptive learning-based difficulty modulation, multilingual support, and integration with real-time learner analytics [9].

EXISTING SYSTEM

- Existing digital platforms may offer question bank repositories; however, they often depend on static, pre-defined questions and lack real-time adaptability. These systems typically:
- Use fixed templates without AI-driven customization.
- Do not leverage Natural Language Processing (NLP) or deep learning to understand context.
- Require manual entry of questions, categorization, and tagging.
- Lack intelligent filtering, formatting, or generation based on syllabus or cognitive difficulty.
- Moreover, there's limited integration of modern **Large Language Models (LLMs)** and **Generative AI**, which are capable of understanding complex educational content and dynamically creating high-quality questions of various types (e.g., MCQs, fill in the blanks, descriptive, etc.).
- As a result, the existing systems fall short in terms of **efficiency, personalization, and automation**, prompting the need for an intelligent, AI-powered solution.

DRAWBACKS

- Manual question paper creation is time-consuming and requires skilled staff.
- Existing tools lack automation, intelligence, and integration for content generation.
- Question generation is prone to human error and inconsistency across subjects.
- Institutions rely on traditional methods without support for dynamic content or updates.

PROPOSED APPROACH

- Utilize Large Language Models (LLMs) to automatically generate high-quality questions from academic content.
- Categorize questions by type (MCQ, descriptive, logical, etc.) and difficulty level using AI prompts.
- Implement topic-wise filtering to generate questions based on book, chapter, and topic selection.
- Automate metadata extraction like subject, medium, and chapter count from uploaded materials.
- Enable easy storage, search, and export of generated questions for examination use.

ADVANTAGES

- **Saves Time and Effort** – Automatically generates questions, reducing manual workload for educators.
- **Consistent Quality** – Ensures question quality and relevance through AI-driven content generation.
- **Customizable Output** – Supports selection of question type, difficulty level, and subject/topic focus.
- **Scalable System** – Can handle large volumes of data and generate questions quickly for any syllabus.
- **Improves Accessibility** – Provides a centralized, easy-to-access platform for managing question banks.

Hardware Requirements

- **Processor:** Intel Core i5 or higher
- **RAM:** Minimum 8 GB (Recommended: 16 GB)
- **Hard Disk:** 256 GB SSD or higher
- **Graphics Card:** Integrated or Dedicated GPU for AI model processing (Optional)
- **Network:** Minimum 1 Mbps internet connection (Recommended: 10 Mbps or higher)

Software Requirements

1. Frontend:

- **Technology Used:** React.js
- **Styling:** Tailwind CSS, Daisy UI
- **Purpose:** Provides a responsive and modern user interface.

2. Backend:

- **Technology Used:** Node.js with Express.js
- **Authentication:** JWT-based authentication
- **Purpose:** Handles API requests, authentication, and data processing.

3. Database:

- **Primary Database:** PostgreSQL (Relational Database for structured data)
- **Secondary Database:** MongoDB (Used for AI Analysis and unstructured data)

Software Requirements

4. AI Integration:

- **Technologies Used:** Gemini, OpenAI ChatGPT API
- **Purpose:**
- AI-powered question generation.
- NLP-based question improvement for better accuracy.

5. Other Tools & Libraries:

- **Axios:** For making API calls.
- **React Router:** For navigation.
- **React Icons & Toastify:** For UI enhancements.

6. Development Tools:

- **IDE:** VS Code
- **Version Control:** Git & GitHub

Modules

1. Frontend Details

- **Technology Used:** React.js
- **Styling:** Tailwind CSS, Daisy UI

Purpose: The frontend is responsible for rendering the user interface and ensuring a smooth user experience. React.js provides a component-based architecture, making the development process modular and efficient. Tailwind CSS is used for styling, allowing for rapid UI development, and Daisy UI provides pre-built components to enhance the design without extra effort.

Key Features:

- User-friendly and modern UI.
- Responsive design for seamless use on mobile and desktop.
- Efficient state management using React Hooks (useState, useEffect, etc.).
- Form validation using React state and error handling

Modules

2. Backend Details

- **Technology Used:** Node.js with Express.js
- **Authentication:** JWT-based authentication

Purpose: The backend acts as the server that processes requests from the frontend, interacts with the database, and manages authentication. JWT (JSON Web Token) ensures secure user authentication and session handling.

Key Features:

- RESTful API development with Express.js.
- Secure user authentication and authorization.
- Middleware for request validation and error handling.
- Data processing and communication between the frontend and database.
- Efficient handling of user sessions using JWT tokens.

Modules

- **3. Database Details**

Primary Database: PostgreSQL (Relational Database)

- Used for storing structured data like user credentials, application settings, and system logs.
- Ensures data integrity and supports SQL queries for efficient data retrieval.
- Relationships between different entities (users, questions, responses) are managed effectively.

Secondary Database: MongoDB (NoSQL Database)

- Used specifically for AI-related data analysis, such as storing AI-generated questions and logs.
- Ideal for handling unstructured or semi-structured data, making it flexible for NLP-based improvements.
- Provides scalability and high-speed data retrieval for AI computations.

Modules

4. AI Integration Details

- **Technologies Used:** AI like Gemini, OpenAI ChatGPT API
- **Purpose:** AI models are integrated to enhance the platform's capabilities by generating questions and improving them using NLP techniques.
- **Key Features:**
 - AI-powered question generation for quizzes, exams, or learning modules.
 - NLP-based question improvement to refine the generated questions and enhance clarity.
 - API communication between the backend and AI models for dynamic content generation.

Modules

5. Other Tools & Libraries

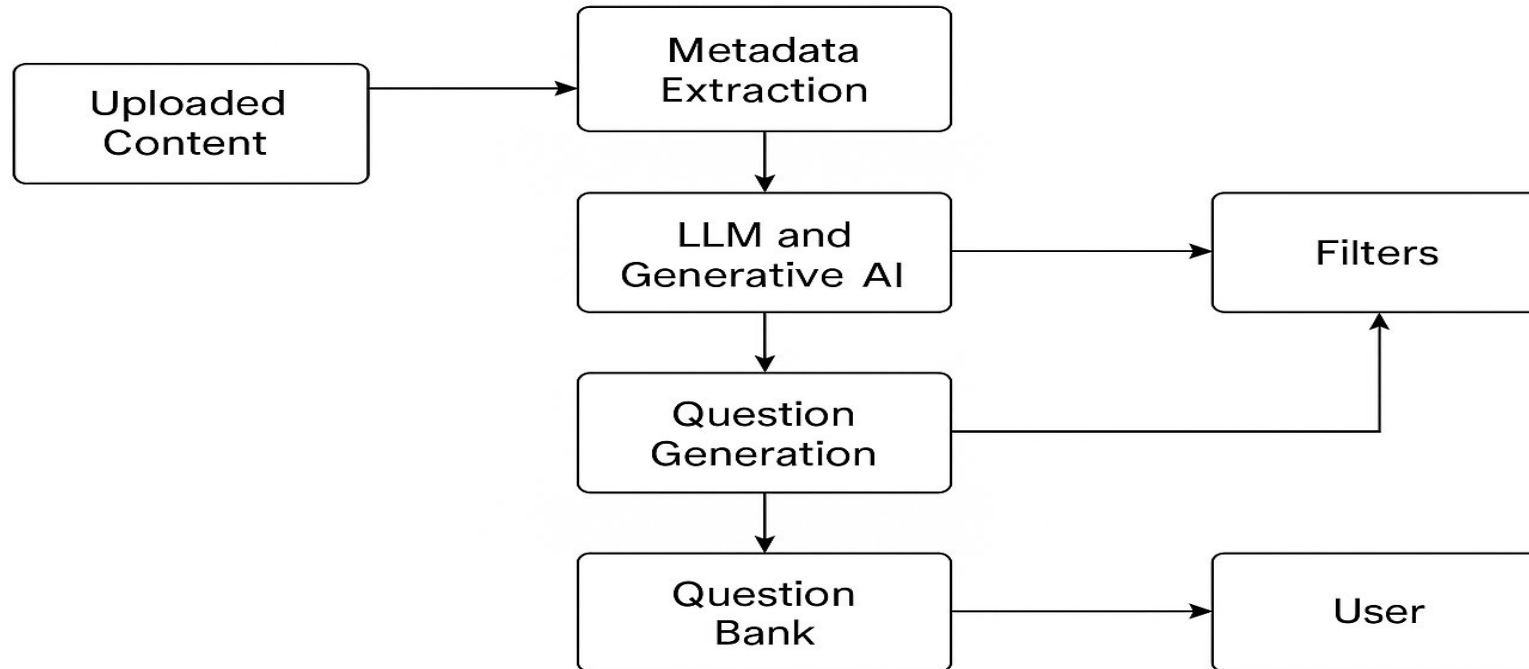
- **Axios:** Used to handle API requests efficiently. It simplifies HTTP calls between the frontend and backend.
- **React Router:** Provides navigation between different pages/components within the React app.
- **React Icons & Toastify:** Enhances UI/UX with icons and real-time notifications for a better user experience.

6. Development Tools

- **IDE:** Visual Studio Code (VS Code) is used for writing and managing code, offering a developer-friendly environment with extensions and debugging tools.
 - **Version Control:** Git & GitHub
 - Git is used for version control, allowing developers to track changes and collaborate efficiently.
- GitHub is used for repository management, making it easy to share and deploy the project.

SYSTEM ARCHITECTURE

System Architecture for Automated Question Bank Generation System Using LLM and Generative AI



CONCLUSION

- The **Automated Question Bank Generation System using LLM (Large Language Models) and Generative AI** represents a significant advancement in educational technology, offering a highly efficient, scalable, and customizable solution for creating question banks. By leveraging the power of LLMs and generative AI, this system automates the process of generating diverse types of questions, including multiple-choice, fill-in-the-blanks, descriptive, and logical reasoning questions, tailored to varying difficulty levels and educational standards.
- This system not only enhances the speed and accuracy of question generation but also allows for greater flexibility in content creation. It can quickly generate questions based on selected books, chapters, and topics, while maintaining a high degree of relevance and alignment with the curriculum. Additionally, the use of advanced AI ensures that the questions are varied, well-structured, and engaging for learners.

FUTURE ENHANCEMENT

- Future enhancements to the Automated Question Bank Generation System could include multi-language support for global accessibility, adaptive learning integration for personalized question difficulty based on student performance, and AI-driven feedback mechanisms to evaluate question quality. It could also feature seamless integration with Learning Management Systems (LMS) for streamlined use in educational platforms, real-time collaboration tools for educators, and multimedia question generation for a more engaging learning experience. Additionally, the system could incorporate bias detection and ethical standards, align with educational frameworks and standards, and provide advanced analytics for real-time performance tracking, ultimately enhancing the system's effectiveness, inclusivity, and adaptability in diverse educational settings.

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Do you have any questions?
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